

Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU

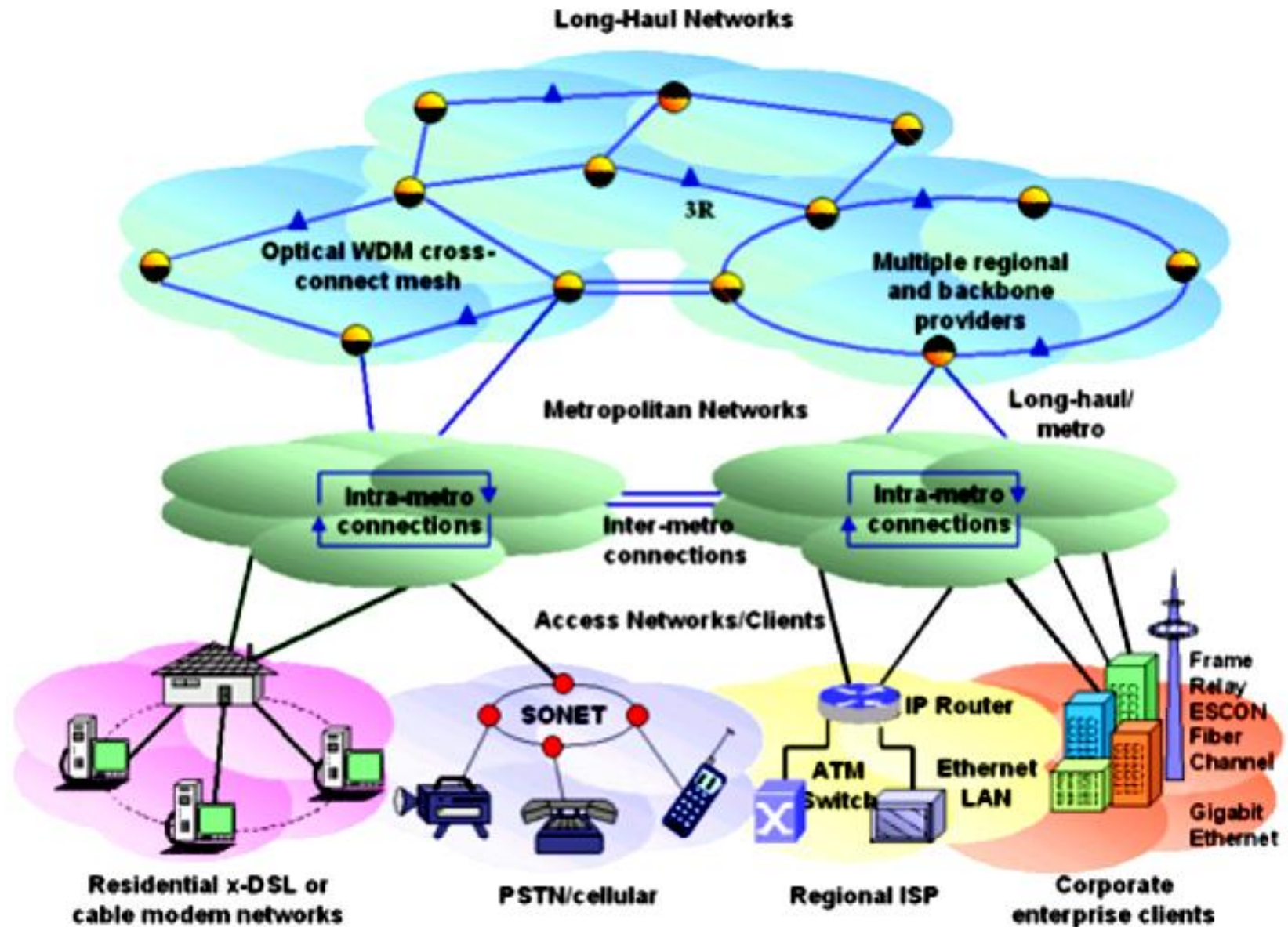
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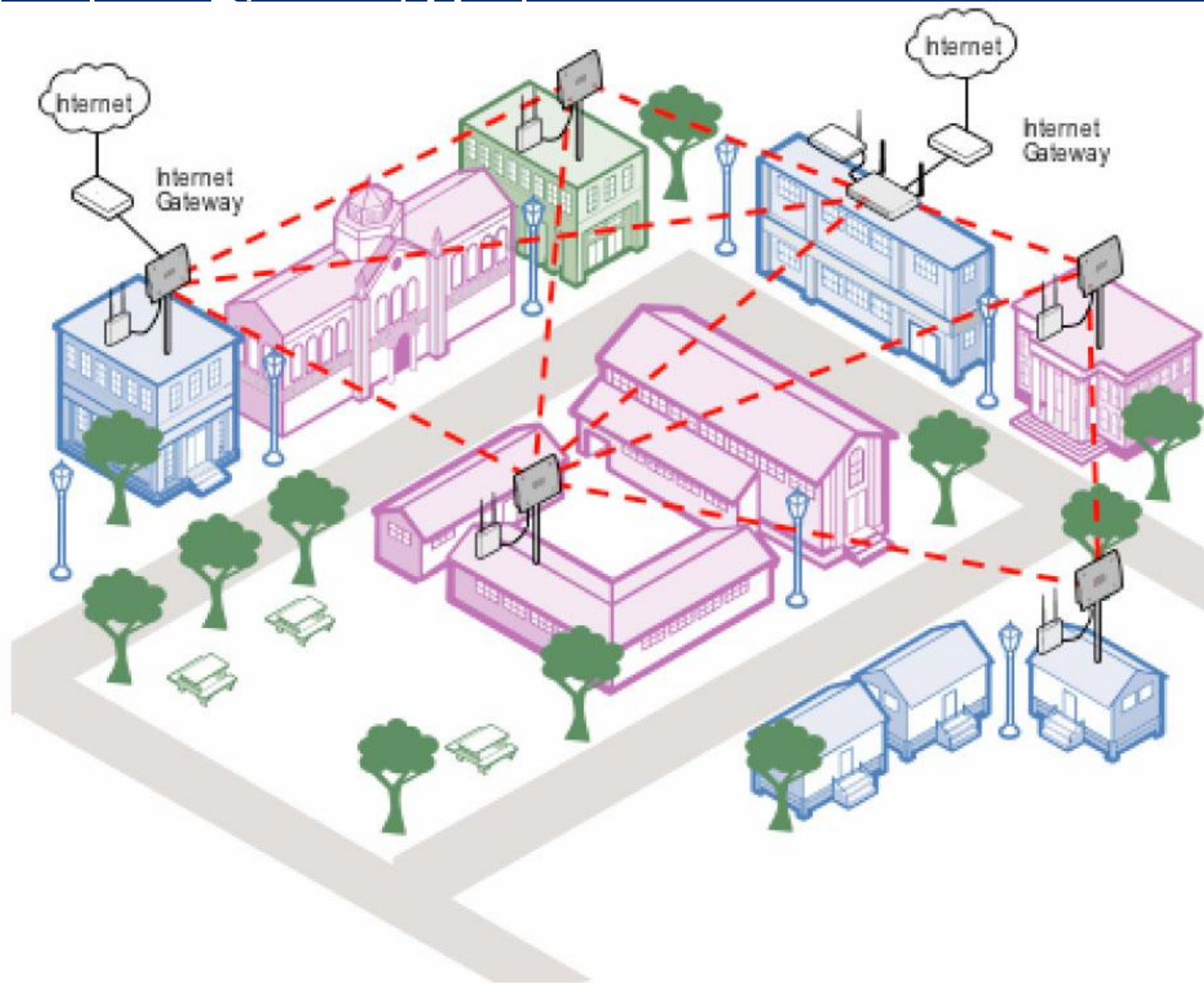
Topic: Metro Networks

Outline

- Introduction
- Design Considerations
 - Compensation Schemes
 - Crosstalk
- Case Studies

A High Level View





Key Requirements

- Multi-protocol, multi-service, and sub-rate support
 - DS-0 (64 Kbits/sec), DS-1 (1.5544 Mbits/sec), DS-3 (44.736 Mbits/sec) and 10/100 fast Ethernet, and traditional ATM and frame relay circuits
 - OC-3 (155 Mbits/sec), OC-12 (622 Mbits/sec), OC-48 (2.5 Gbits/sec), OC-192 (10 Gbits/sec)
- Optical transparency
 - Bit rate and format independent. WDM might be the key technology
- Scalability
 - Topological flexibility is required
 - WDM-based rings by using re-configurable OADM are proposed
 - Mesh network topology is another alternative
- Multiple survivability options
 - Protection: dedicated/shared protection, pre-provisioned recovery
 - Restoration: dynamic signal recovery
- Low cost

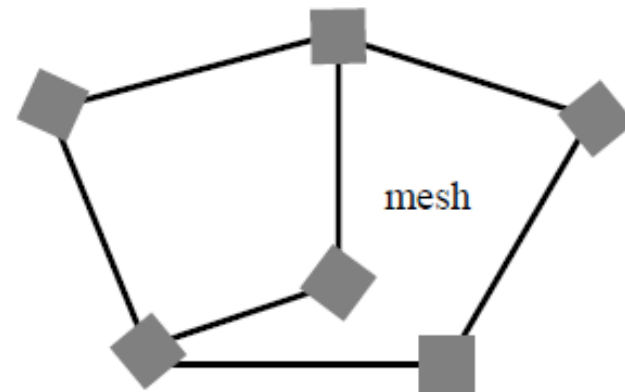
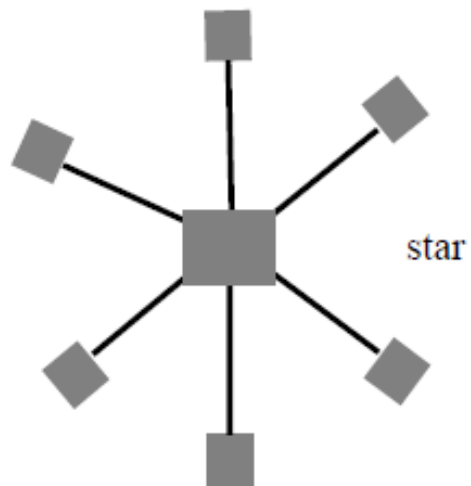
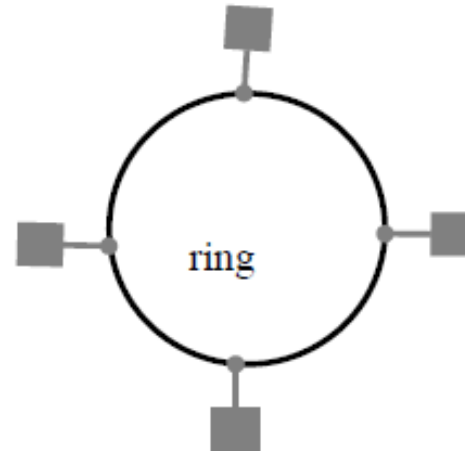
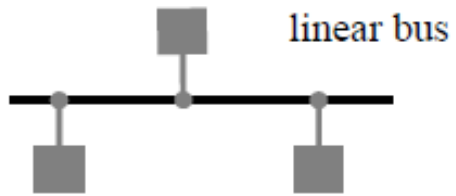
Features of Long haul v.s. Metro

	Long-haul	Metro
Transmission distance	Hundreds to thousands of km	Typically up to 100 km
Traffic channels	2.5 Gbps, 10 Gbps, and 40 Gbps	Highly diverse, SONET-like channels coexist with Fiber and Gigabit Ethernet channels
Fiber topology	Linear	Ring or mesh
Traffic topology	Linear	Hub-and-spoke
Traffic protection	Implemented by OXC, DCC, or upper layer protocol. Transmission system itself does not provide protection	Transmission system must provide protection or support client layer protection mechanisms
Migration	Limited to additional channels	Addition of both capacity and new ring sites
Cost	Low cost is an advantage	Must be low enough to compete with alternative technologies and the provision of additional fiber
Footprint	Small size is an advantage	Small size is crucial

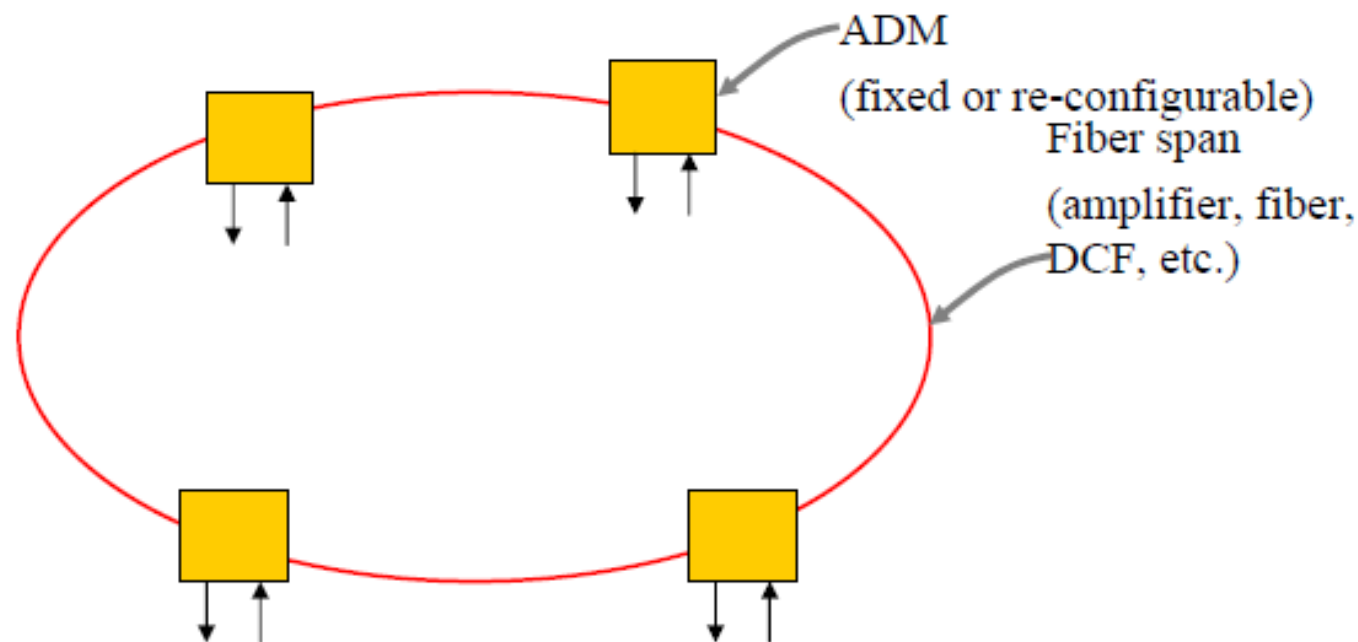
Components Selection

- Transmitters
 - Externally or directly modulated lasers
- Wavelength add/drop Multiplexers
 - Fixed or re-configurable
- Switches and optical cross-connects (OXC)
 - Fiber switching
 - Wavelength switching
 - Wavelength conversion
- Optical amplifiers
 - Lower noise and flatter, stable gain is desired
- Fibers
- Receivers
 - Photo-detectors: PIN, APD (avalanche photodiode)

Topologies



Basic Ring



- Consider a typical WDM metro ring at four OADMs, as shown in the above figure.

Typical Losses

<i>Symbol</i>	<i>Description</i>	<i>Loss value</i>
L_{mon}	Tap loss for monitoring	0.2 dB
L_{conn}	Loss of a connector	0.25 dB
L_{splice}	Loss of a splice	0.15 dB
L_{span}	Loss of the fiber span	0.25 dB/km
L_{mux}	Loss of a multiplexer	4 dB
L_{demux}	Loss of a demultiplexer	4 dB
L_{foadm}	Loss of a fixed OADM	2 dB
L_{roadm}	Loss of a re-configurable OADM	10 dB

Direct or
External
Modulation

<i>Symbol</i>	<i>Description</i>	<i>Optical Power Sensitivity</i>
$P_{T-DM @ 2.5}$	2.5 Gbps/DM output power	3 dBm
$P_{T-EM @ 10}$	10 Gbps/EM output power	0 dBm
$P_{R-sen @ 2.5}$	2.5 Gbps pin diode receiver sensitivity	-23 dBm
$P_{R-sen @ 10}$	10 Gbps pin diode receiver sensitivity	-16 dBm

DM, EM ???

$$L_{allowable} = P_T - P_{R-sens}$$

$$L_{node} = 2L_{splice} + 6L_{conn} + L_{f/r oadm}$$

An example

- Without amplification, the allowable loss is given by

$$L_{allowable} = 3 - (-23) = 26 \text{ dB}$$

If we consider that the carriers require metro rings with circumferences of up to 200 km and 4 nodes, we conclude that *this cannot be achieved without amplifiers*.

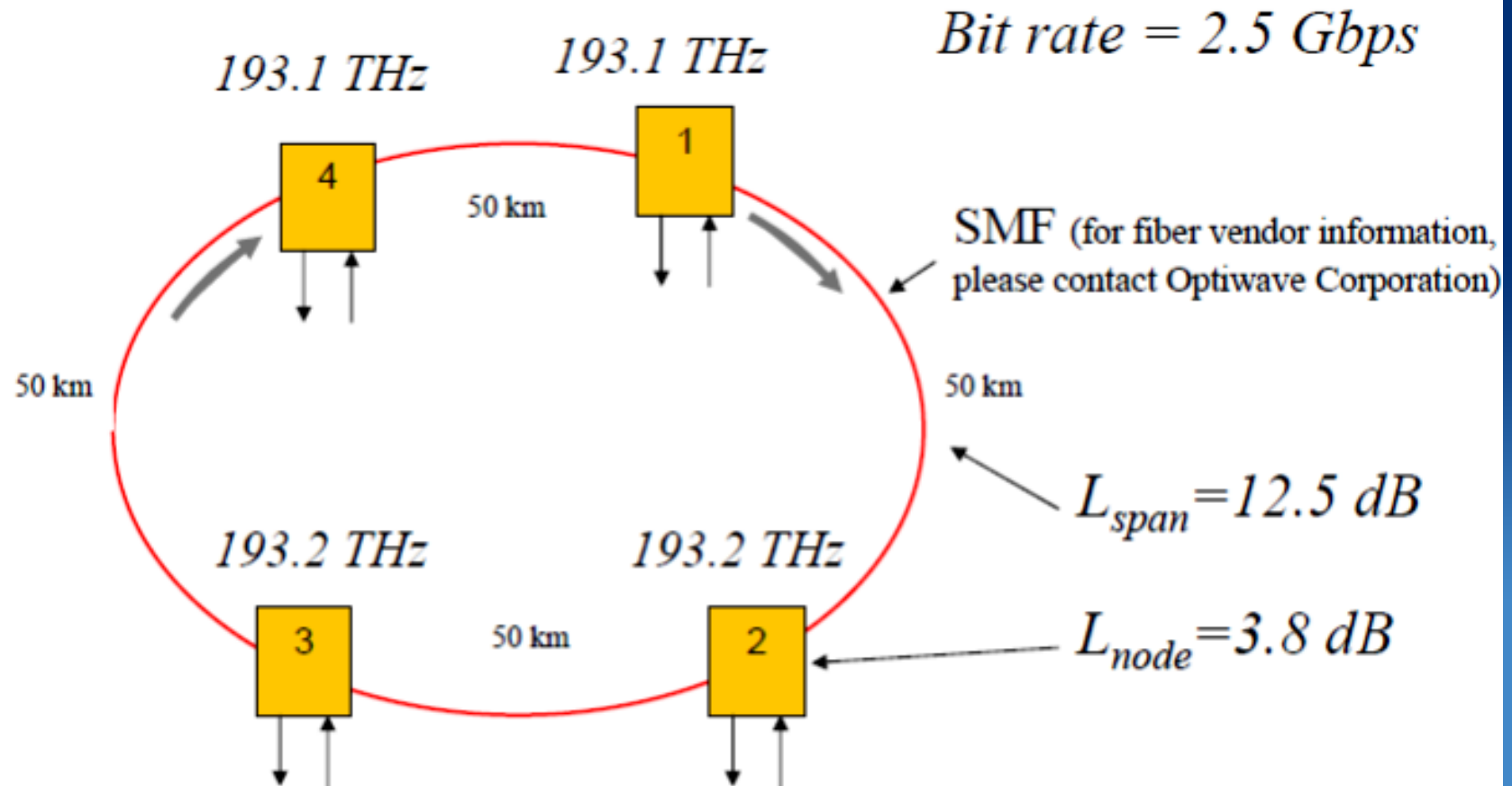
- Total node loss (including two splices and six connectors) is given by

$$L_{node} = 2 \times 0.15 + 6 \times 0.25 + 2 = 3.8 \text{ dB}$$

- Span loss will be $L_{span50km} = 50 \times 0.25 = 12.5 \text{ dB}$

- Total loss will be $L_{span200km} = 12.5 \times 4 + 3.8 \times 4 = 65.2 \text{ dB}$

Without Amplifier Option



$$P_{required} = 65.2 - 23 - 3 = 39.2 \text{ dBm/channel}$$

- At this power level, we need to consider fiber nonlinearity.

Self Phase Modulation

- SPM effects are negligible when $P_0 < \alpha/\gamma$
- For the fiber we used $\gamma \approx 1.5 \text{ W}^{-1}\text{km}^{-1}$
- SPM effects can be negligible when the peak power is below *166 mW or 18 dBm average power*
- If you use N_A amplifiers along the link, the criteria becomes $P_0 < \alpha/(\gamma N_A)$. If you use two amplifiers along the link, maximum allowable power before the nonlinearity becomes effective drops by 3 dB.
- Using dispersion management by using DCF can reduce SPM

XPM: Cross Phase Modulation and FWM

- As a rough estimate, the channel power is restricted with $P_{ch} < \alpha / [\gamma (2N_{ch} - 1)]$

Here, N_{ch} is the number of channels

- - FWM efficiency depends on signal power, channel spacing, and dispersion
 - If the GVD of the fiber is relatively high $|\beta_2| > 5 \text{ ps}^2 / \text{km}$, the FWM efficiency factor almost vanishes for a typical channel spacing of 50 GHz or higher
 - If the channel is close to zero dispersion wavelength of the fiber, considerably high power can be transferred to FWM components.
 - To reduce the effect of FWM to the system performance, you can use either uneven channel spacing or use dispersion-management technique

Stimulated Raman and Brillouin

- The Raman threshold for a single channel system is given by

$$16.4 \text{ dBm} / \text{km}$$

- Threshold level depends on source line-width, effective core area, and effective fiber length.

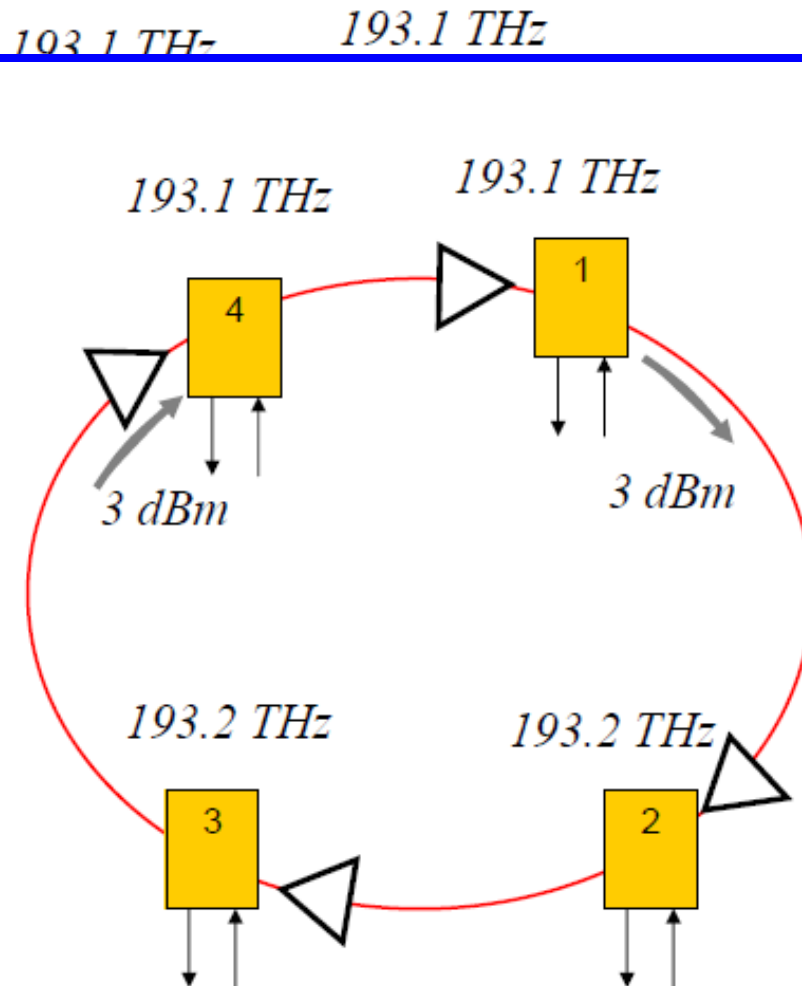
$$P_{th} \approx \frac{21 A_{eff}}{g_B L_{eff}}$$

- Typical value for g_B is around $5 \times 10^{-11} \text{ m/W}$
- The threshold value also depends on modulation format and duration of pulse.
- Some values:
 - 9 dBm for CW light
 - 12 dBm for externally modulated transmitter
 - >18 dBm for externally modulated transmitter with source wavelength dither
- Generally speaking, SBS has little effect on system performance

With Amplifiers: Lumped v.s. Per node

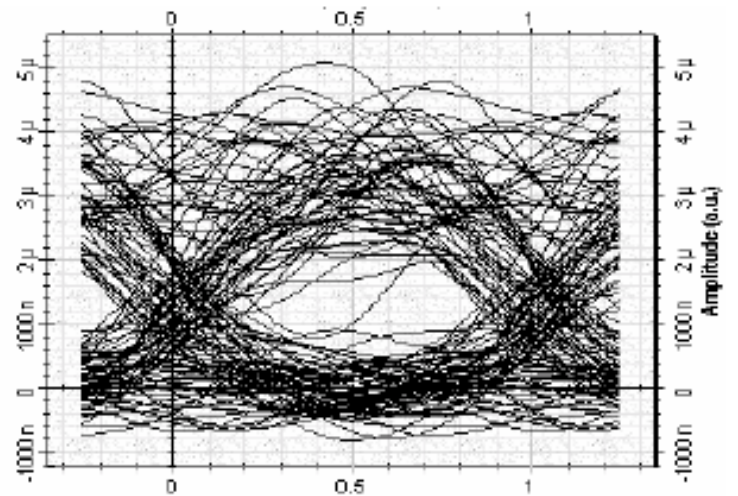
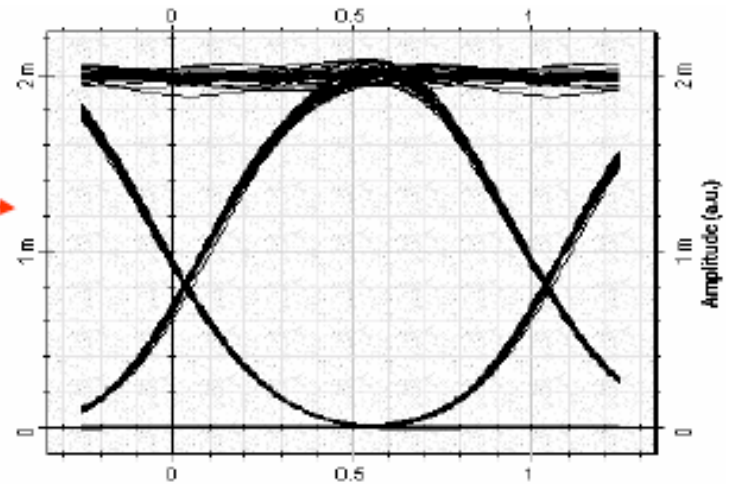
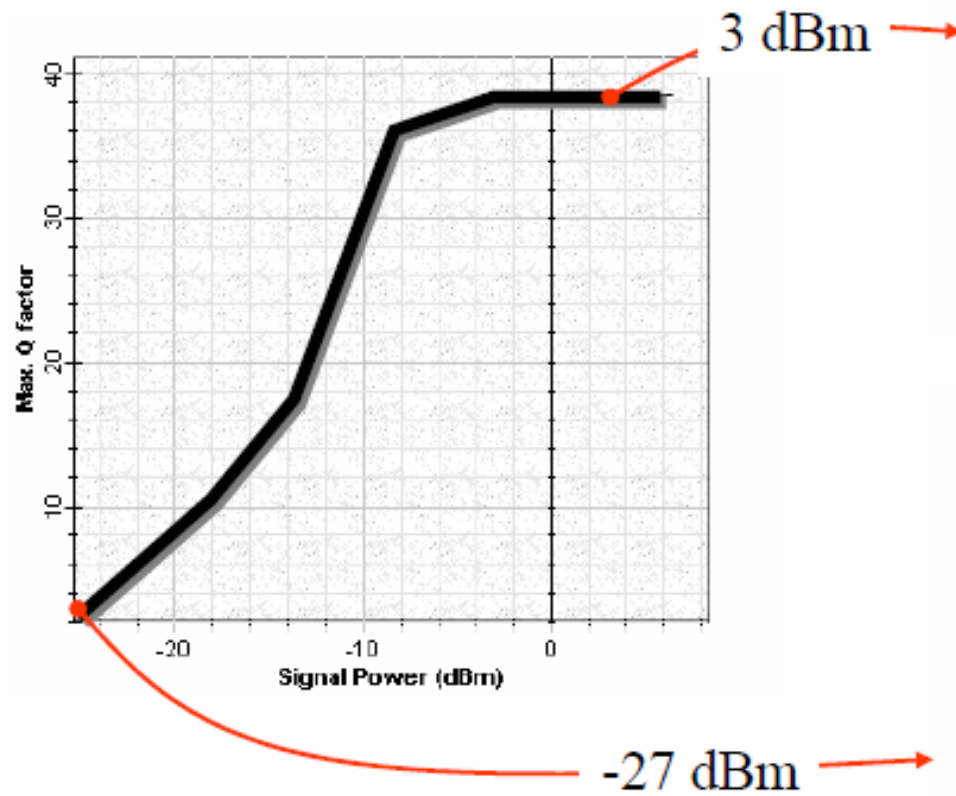
Alternative 1: “Lumped” amplification

- 25 dB is higher than the



- Each amplifier will match the loss of one node and the span preceding it
- Brings the *topological flexibility* (a new node can be added easily)
- *Nonlinearity* is not an issue
- Requires *low cost-compact* optical amplifiers

Simulation results

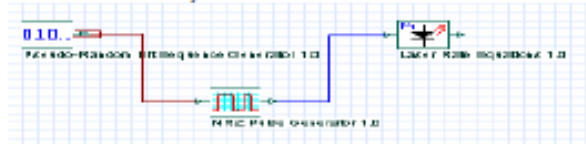


Migrating to 10Gb/s

Loss compensation

GVD limited transmission distance

- Directly modulated DFB lasers



For $D=16 \text{ ps}/(\text{km}\cdot\text{nm})$
at 2.5 Gbps, $L \approx 42 \text{ km}$

- Externally modulated source

For $D=16 \text{ ps}/(\text{km}\cdot\text{nm})$
at 2.5 Gbps, $L \approx 500 \text{ km}$

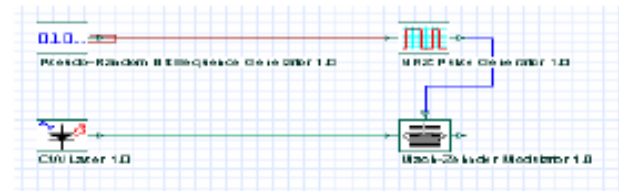
at 10 Gbps, $L \approx 30 \text{ km}$

Dispersion compensation is required

$$L < \frac{1}{4B|D|\sigma_\lambda}$$

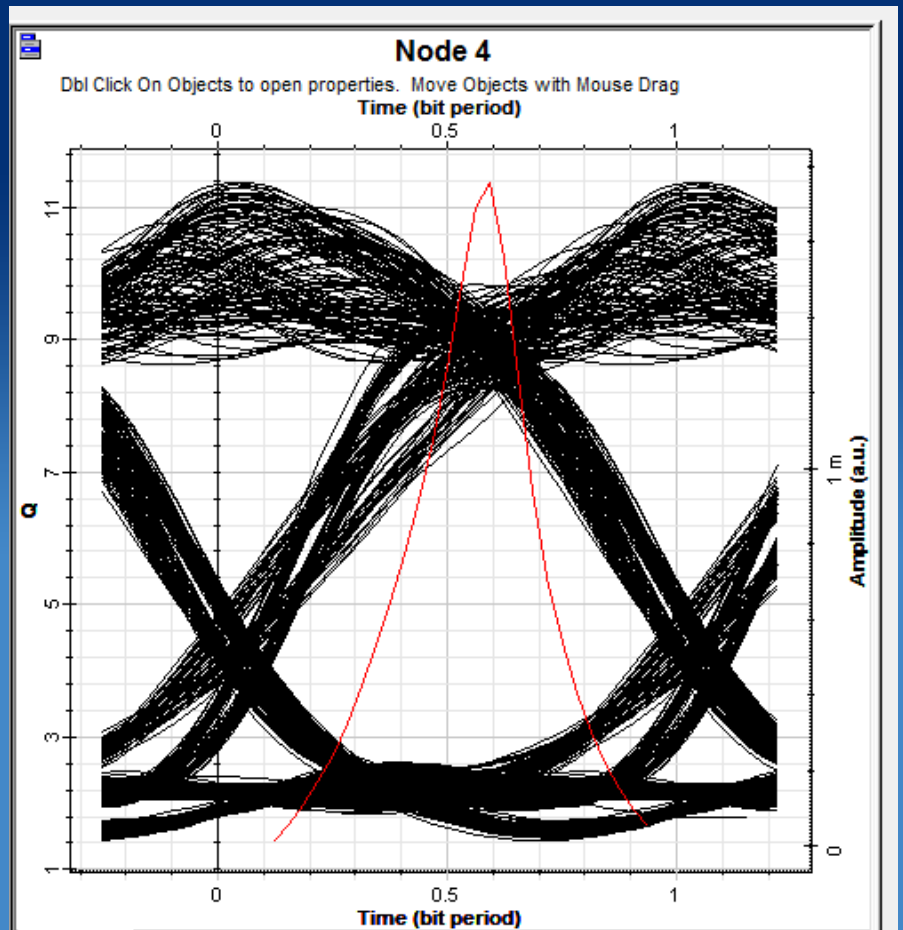
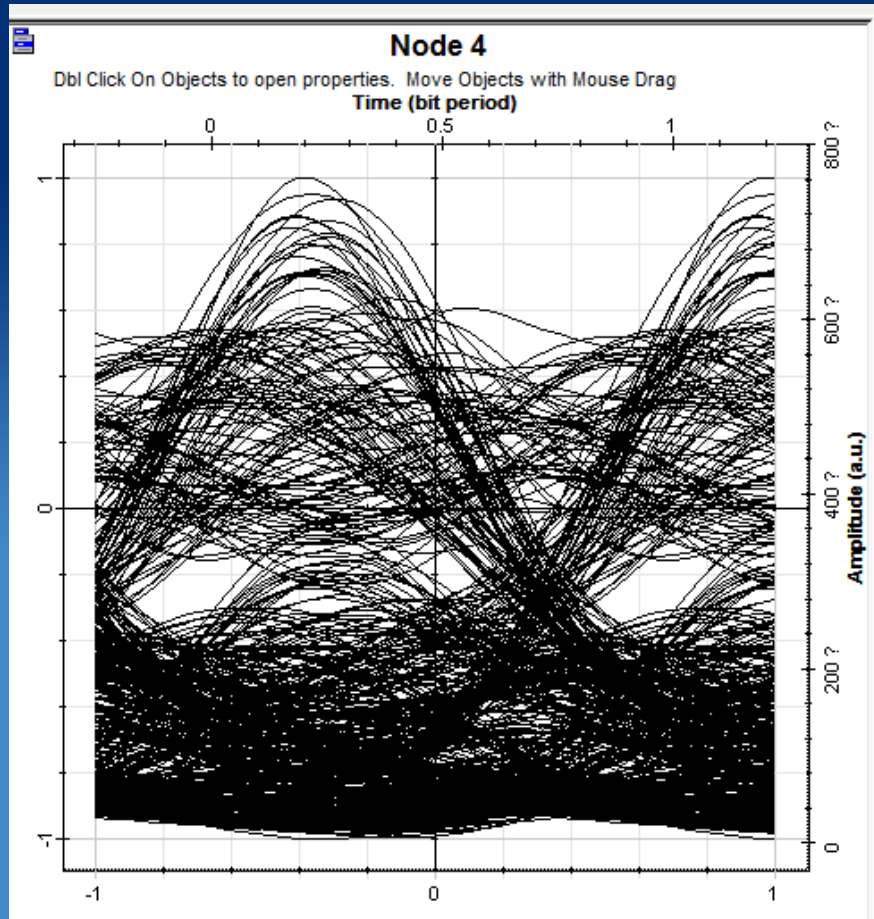
root-mean-square spectral width, a typical value is around 0.15 nm

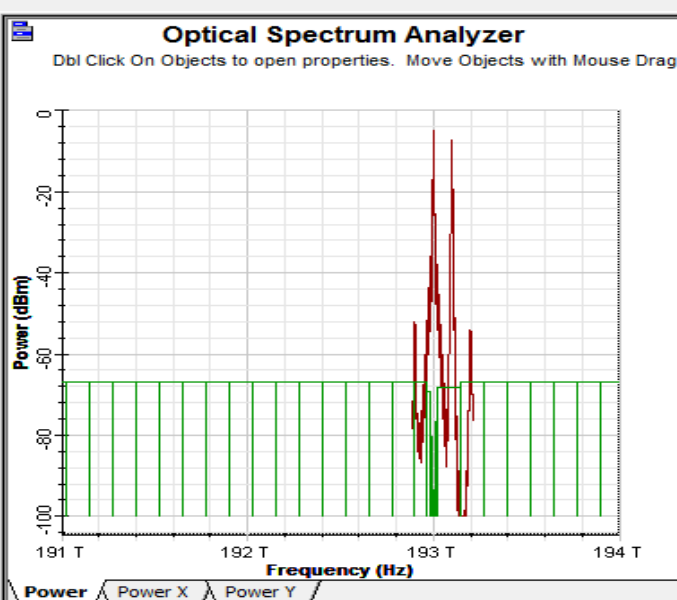
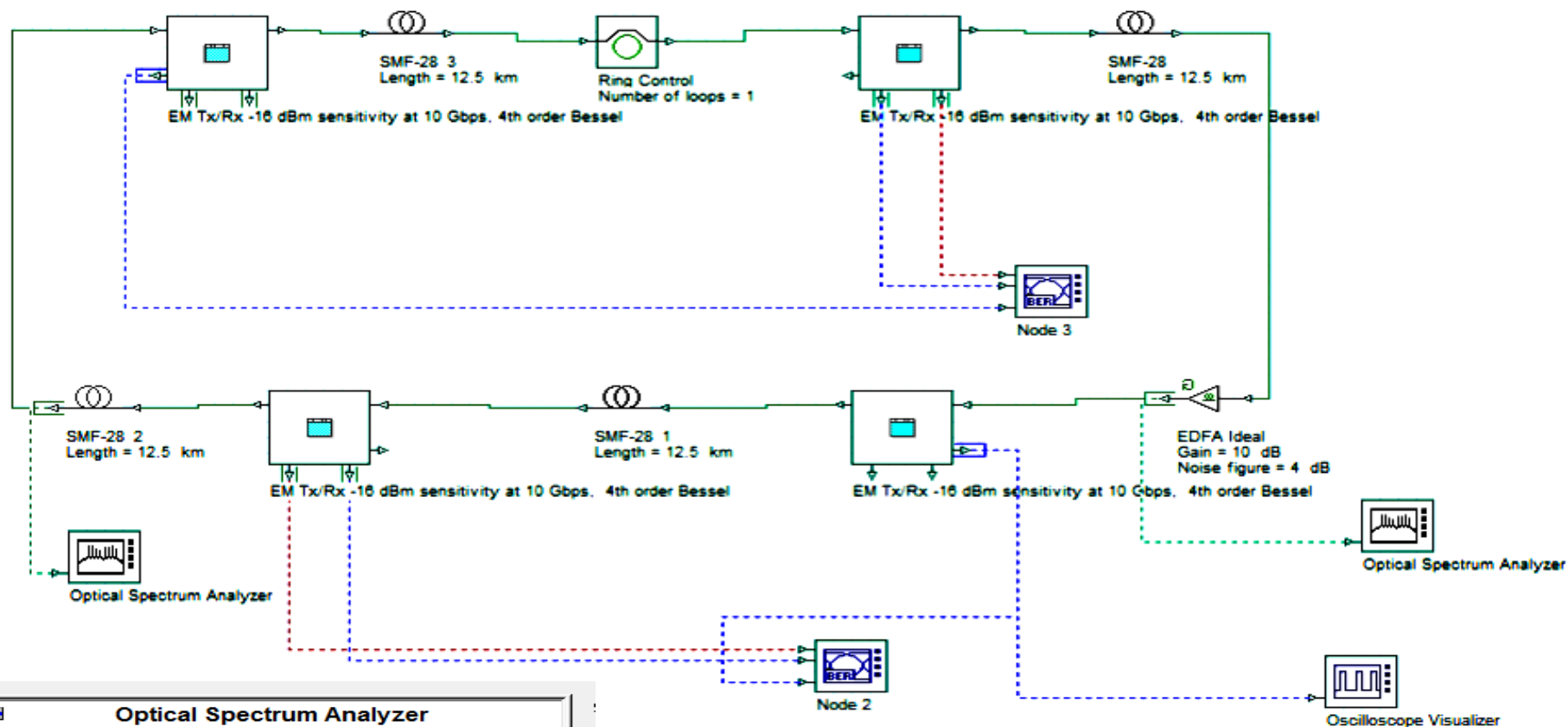
$$L < \frac{2\pi c}{16|D|\lambda^2 B^2}$$



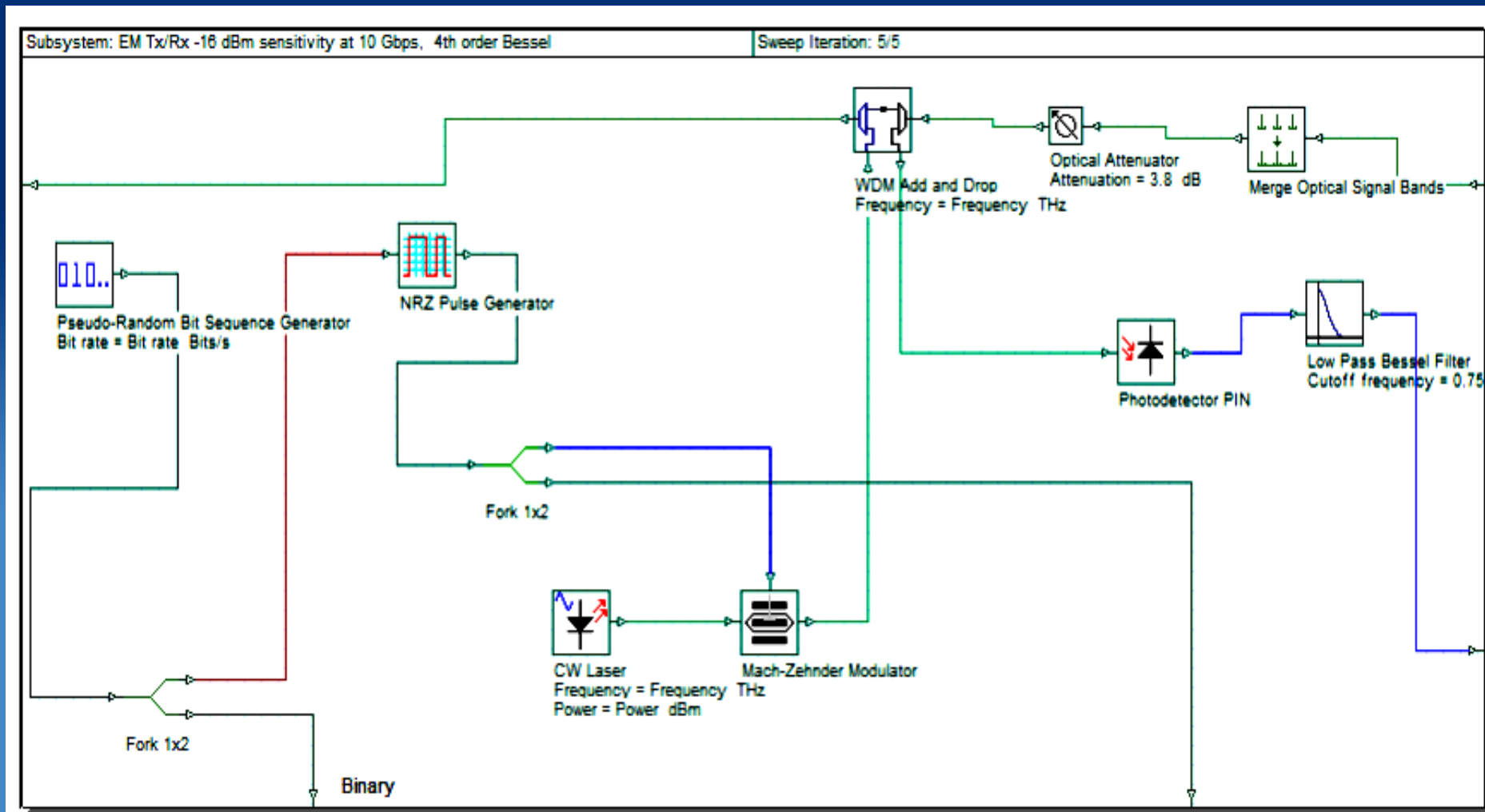
Lumped DCF v.s. at each node

Ring Circumference: 200km



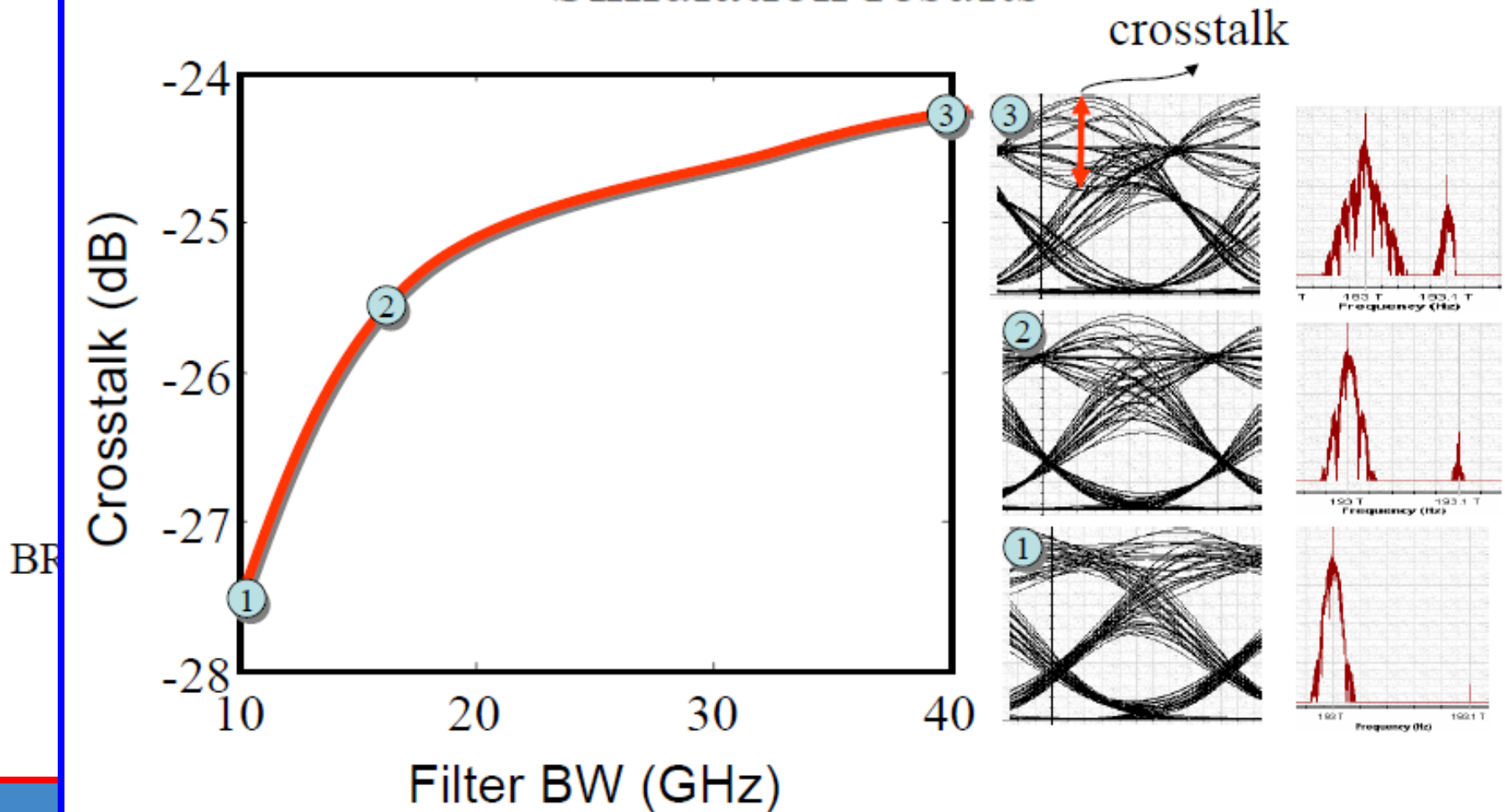


Look inside an EM Tx Rx



Example: Inter-channel Crosstalk

Simulation results



•Tim Gyselings et. al, "Crosstalk analysis of multi-wavelength optical cross connects", J. Light. Tech. 17, pp. 1273, 1999.

Interchannel Crosstalk

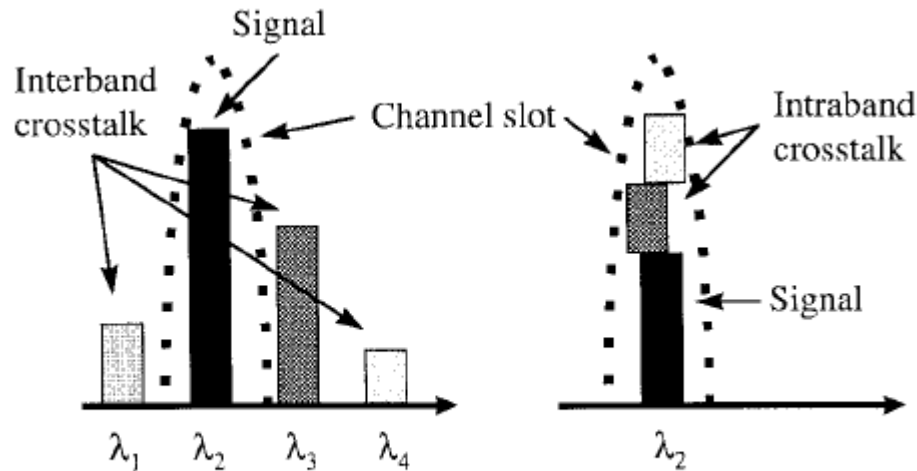


Fig. 5. Interband and intraband crosstalk.

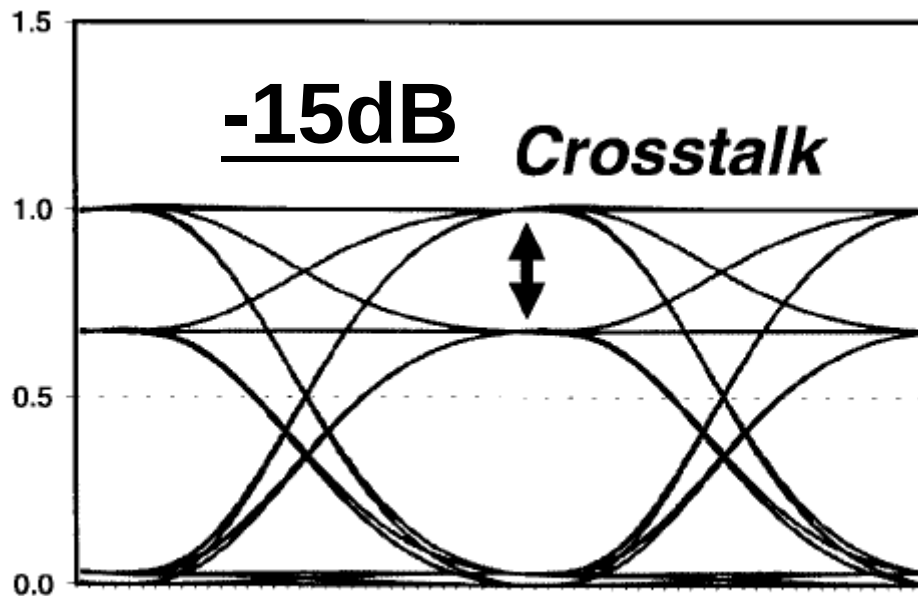
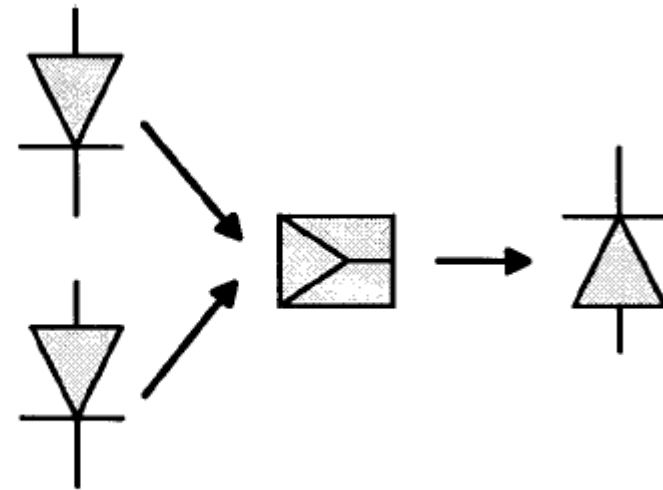
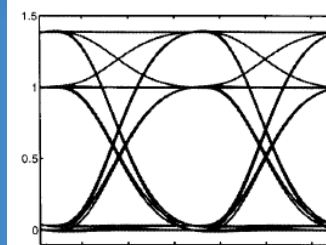


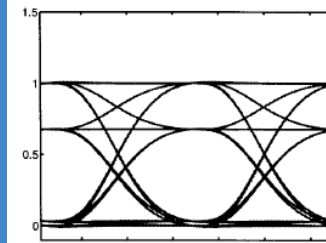
Fig. 8. Definition of crosstalk.



$$P_{\text{out}} = P_1 + P_2 + 2\sqrt{P_1 P_2} \cos[(\omega_1 - \omega_2)t + \Phi_1(t) - \Phi_2(t) + \Theta_1 - \Theta_2]$$



(a)

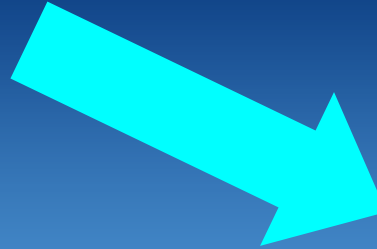
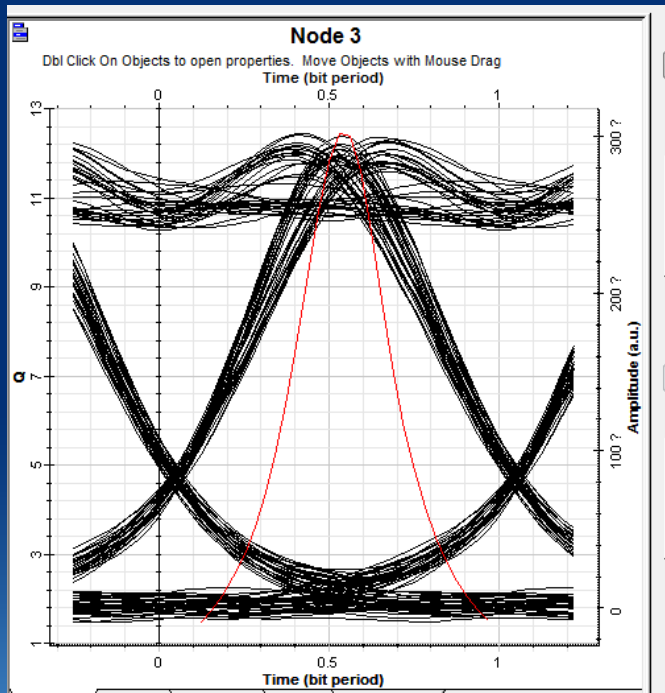


(b)

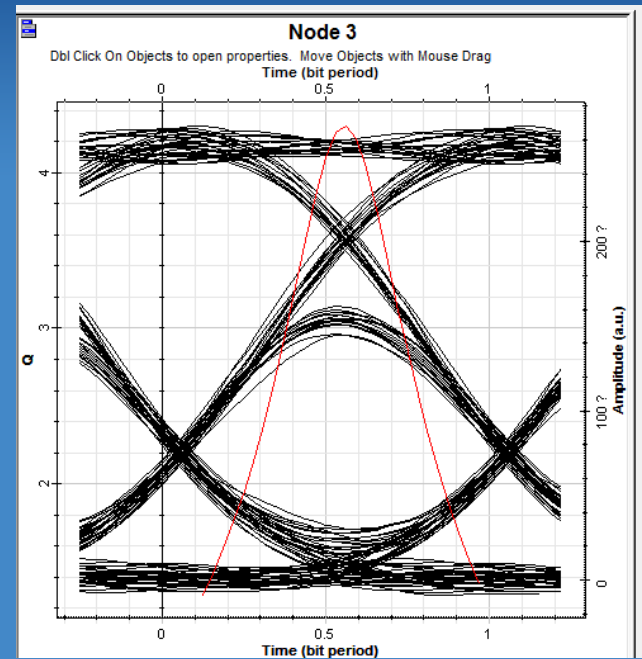
Positive

Negative

10GHz



40GHz



Today's Task:

- For a 10Gb/s system
 - Q1: at what ring circumference will lumped DCF compensation be close to the compensation per node?
 - Q2: Observe Crosstalk with two adjacent channels
- upgrade to 40Gb/s system

